Supplementary Material

Relevant, But Not Useful: Prospective Utility in Agentic Search

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This supplement provides implementation details, the preregistration and provenance protocol, the agent compliance audit, exact label and feature definitions, and detailed diagnostic and robustness results referenced in the main paper. Section 1 details the agent and retrieval stack, including the stronger relevance selectors; Section 2 the candidate pool, selector configurations, and the choice-space consistency audit; Section 3 the preregistration chain and analysis provenance; Section 4 the cross-family compliance audit and post-hoc replication; Section 5 outcome labels and evaluation targets; Section 6 observable state, candidate, and continuation features; Section 7 additional diagnostic results; Section 8 robustness and sensitivity results; and Section 9 reproducibility information.

1 Agent and Retrieval Stack

Search agent. The agent follows a ReAct-style loop: at each turn it emits either a search action (a query wrapped in `<search>` tags) or a final answer (wrapped in `<answer>` tags). Each search returns the top-3 documents (title and abstract text) from the fixed corpus, appended to the context as an observation block. Episodes allow at most 4 search turns; confirmatory continuations use greedy decoding (temperature 0) with stop sequences on the action delimiters and on generation of an observation-opening tag, so that the model cannot hallucinate its own retrieval results. Unclosed action tags at end-of-generation are parsed conservatively (an unclosed `<search>`/`<answer>` span is accepted only when it is unambiguous; occurrences are counted and audited). A preregistered parser audit found no evidence that these safeguards caused the observed trajectory differences: parser versions produced 94% identical trajectories, compared with 76% under a same-code control rerun in the same production environment, and no safeguard-specific mechanism activated in the confirmatory runs. Additional post-hoc robustness analyses are described in Secs. 7–8.

Continuation horizon. Intervention replays do not receive a fresh search budget. The replayed episode inherits the number of search calls already consumed by the frozen prefix, so a state reached after k searches has at most $4 - k$ search turns remaining. On the HotpotQA lockbox intervention arms, the frozen states occur after 1/2/3/4 consumed searches in 2,800/1,895/656/136 cases, respectively. The realized continuation uses 0/1/2/3 additional searches in 2,989/1,739/608/151 cases, for an average of 0.62 additional search steps. When the global search budget is exhausted, the agent is instructed to produce a final answer immediately.

Corpus. All settings use the official HotpotQA `enwiki-20171001` abstract corpus (5,233,235 documents). 2WikiMultiHopQA and MuSiQue are evaluated against the same fixed corpus; annotated

supporting documents are covered for 92.7% of 2WikiMultiHopQA and 98.0% of MuSiQue validation questions, and confirmatory 2WikiMultiHopQA evaluation is restricted to covered questions (main paper, Sec. 4.1).

Retrievers. Lexical retrieval uses BM25 (`bm25s`). Dense retrieval uses `intfloat/e5-base-v2` with brute-force inner product over the full corpus (fp32). The primary relevance baseline is the MS MARCO cross-encoder `cross-encoder/ms-marco-MiniLM-L6-v2`, scoring each candidate's full title and passage text against the current query q_t in the original confirmatory configuration. All retriever configurations are fixed across the five evaluation settings.

Stronger relevance selectors. We additionally evaluate `BAAI/bge-reranker-v2-m3` using either the current query q_t or a state-aware representation containing q_0, q_t, and consumed evidence. We also evaluate a Qwen2.5-7B listwise relevance ranker over the observable state. These models are used only as relevance selectors; they do not observe intervention outcomes. For BGE rescoring, each document is represented as its title followed by the first 600 characters of passage text (tokenizer maximum length 512); the exact full-pool rescoring audit uses this identical configuration. The MiniLM CE baseline scores the full passage text (title and text, tokenizer maximum length 384) throughout.

2 Candidate Pool and Relevance Selectors

For every eligible state, the retrieval pool $\mathcal{C}_t$ is the union of BM25 top-20 lists under q_t, q_0, and up to six already-seen document titles (entity-style queries), plus dense top-20 lists under q_t and q_0, excluding all previously consumed documents (109.1 ± 27.7 documents per state in the primary evaluation). Up to eight typed candidates are drawn per state:

- **bm25_top** — BM25 top-1 under q_t;
- **dense_top** — dense top-1 under q_t;
- **ce_top** — the highest-ranked CE candidate not already retained by an earlier standard family; the full-pool CE argmax is always retained in $\mathcal{D}_t$ and defines the relevance-selection baseline, as detailed below;
- **mid_rel** — a BM25 rank 9–16 document under q_t;
- **novelty** — in the pool but outside BM25 top-20 under q_t and outside dense top-5 (low current-query relevance, reached via q_0/dense-tail expansion);
- **redundancy** — high q_t relevance (BM25 top-10) with the highest lexical (Jaccard) overlap against already-seen evidence (retained only if overlap > 0.05);
- **entity_bridge** — reached only via entity-title queries and absent from the BM25 ranking under q_t;
- **random** — a random remaining pool member.

Consistency of the CE baseline. Candidates are drawn without replacement, and if the pool-level CE argmax has already been taken as **bm25_top** or **dense_top**, it remains in $\mathcal{D}_t$ under that label while **ce_top** takes the next-ranked document. Consequently, $\arg\max_{d \in \mathcal{D}_t} R_{\mathrm{CE}}(q_t, d)$ and $\arg\max_{d \in \mathcal{C}_t} R_{\mathrm{CE}}(q_t, d)$ select the same document, and the main paper's d_{CE} (defined over $\mathcal{D}_t$) equals the pool-level CE selection. We verified this empirically: in all 3,881 evaluated states across the five settings, the pool-level CE argmax is present in $\mathcal{D}_t$ (zero exceptions).

Standard-retrieval-only analyses restrict the oracle to {**bm25_top**, **dense_top**, **ce_top**}.

2.1 Choice-Space Consistency Audit

Full-pool argmax retention. Table 1 reports, for each relevance selector, the fraction of evaluated states whose full-pool argmax over $\mathcal{C}_t$ is already contained in the retained intervention set $\mathcal{D}_t$. By construction this holds for the CE baseline; for the stronger selectors the full-pool argmax is frequently absent from $\mathcal{D}_t$.

Table 1: Fraction of states whose full-pool selector argmax is retained in $\mathcal{D}_t$.

Setting	CE	BGE (q_t)	BGE-State
HotpotQA dev	1.000	0.668	0.366
HotpotQA lockbox	1.000	0.650	0.318
2Wiki	1.000	0.578	0.249
Qwen2.5-14B	1.000	0.706	0.372
MuSiQue	1.000*	0.602	0.264

*The original logged candidate pools retain the CE full-pool argmax in all 3,881 evaluated states. A single MuSiQue mismatch appears only when reconstructing the pool for the post-hoc audit, due to a dense-retrieval boundary-rank reconstruction difference; it is not a mismatch in the original intervention pool.

$\mathcal{D}_t$-space relevance quality. Table 2 reports the support-document hit rate of each selector's choice within $\mathcal{D}_t$, computed only among states for which $\mathcal{D}_t$ contains at least one annotated support document.

Table 2: Support hit rate of selector choices within $\mathcal{D}_t$, computed only among states for which $\mathcal{D}_t$ contains at least one annotated support document. The final column reports the BGE-State − CE difference with question-level 95% confidence intervals.

Setting	CE	BGE	BGE-State	BGE-State − CE
HotpotQA dev	0.753	0.846	0.900	+0.147 [0.074, 0.223]
HotpotQA lockbox	0.719	0.801	0.903	+0.184 [0.100, 0.268]
2Wiki	0.707	0.796	0.790	+0.082 [0.007, 0.158]
Qwen2.5-14B	0.749	0.819	0.852	+0.103 [0.038, 0.172]
MuSiQue	0.555	0.671	0.685	+0.130 [0.044, 0.216]

Rollout of missing full-pool argmax candidates. Because BGE and BGE-State full-pool argmax candidates are frequently absent from $\mathcal{D}_t$, we conducted an explicitly post-hoc sensitivity analysis after the lockbox had been opened. We rolled out every missing full-pool BGE and BGE-State argmax under the same frozen greedy continuation policy, yielding 3,861 additional intervention arms and no missing outcomes. Table 3 reports the resulting downstream comparison.

Table 3: Post-hoc full-pool choice-space sensitivity. Entries report downstream F1 differences between each selector's actual full-pool argmax and CE selection, with question-level 95% confidence intervals.

Setting	**BGE(full)−CE**	**BGE-State(full)−CE**
HotpotQA dev	−.019 [−.042, .004]	+.012 [−.016, .043]
HotpotQA lockbox	+.008 [−.010, .026]	+.024 [−.003, .051]
2Wiki	+.010 [−.006, .025]	+.018 [−.007, .043]
Qwen2.5-14B	−.010 [−.029, .010]	−.009 [−.034, .016]
MuSiQue	−.001 [−.014, .012]	−.000 [−.017, .017]

All 10 confidence intervals include zero. Thus, even when BGE and BGE-State are allowed to select their actual full-pool argmax candidates, their large relevance advantage does not yield a statistically detectable downstream gain over CE. This analysis is post hoc and is reported only as a choice-space sensitivity check.

3 Preregistration, Lockbox, and Analysis Provenance

Preregistration chain. All confirmatory decisions and pre-specified diagnostic analyses were fixed in a version-controlled design-revisions document *before* the corresponding data were inspected, including: the primary endpoints and lockbox analysis freeze list; the replication protocols for 2WikiMultiHopQA, the cross-scale agent, and MuSiQue; and additional robustness and diagnostic analyses (cross-fitted oracle, question-clustered inference, pool-robustness, and effect-size decomposition). No acceptance criterion was relaxed after unblinding at any point; criteria that failed remain recorded as failures together with the adjudication that followed.

Single-open confirmatory lockbox. The held-out HotpotQA lockbox was sealed before predictor development: question IDs were selected and committed, all models used for confirmatory evaluation were frozen to disk, and SHA-256 hashes of the frozen artifacts and design document were recorded before opening. The preregistered confirmatory lockbox evaluation was executed once. After all confirmatory analyses were completed, we conducted explicitly post-hoc robustness and diagnostic analyses on the already opened lockbox. These analyses introduced no new confirmatory hypothesis testing and are reported separately from the preregistered results.

Confirmatory versus post-hoc analyses. Confirmatory (frozen before the corresponding data were inspected): the primary Oracle–CE endpoint and opportunity rates, the lockbox evaluation, the representative non-enriched strata, and the split-replay and noise checks. Explicitly post-hoc and exploratory: the choice-space full-pool rollout expansion, winner characterization, the within-state outcome-permutation analysis, budgeted lookahead diagnostics, the post-rollout continuation scorer, sampled-decoding sensitivity, the Phi-4 cross-family replication, the forced-answer myopic-utility analysis, and alternative control-matching specifications.

Measurement-integrity control. A preregistered harness audit compared agent behavior across parser versions. The strict no-op criterion (99% identical trajectories) was *not* met (94%); a preregistered same-code control (re-running the identical legacy code twice under production conditions) yielded 76% identical trajectories, attributing the discrepancy to serving-level nondeterminism, and zero activations of the new parsing mechanisms were observed in confirmatory runs. We therefore

report "no evidence of a behavioral effect" rather than a proven no-op, and record the original criterion as failed.

4 Cross-Family Compliance Audit

Cross-family agent replications were subject to preregistered tool-protocol compliance criteria evaluated on a 30-question smoke set before any confirmatory run (malformed action rate $\leq 5\%$, valid answer rate, token-limit truncation rate, and zero answer leakage). Table 4 summarizes the audit. Only Qwen2.5-14B passed this preregistered audit; Mistral-7B-v0.3 and Llama-3.1-8B were excluded *before* confirmatory evaluation, and the preregistration prohibits re-measuring an excluded model. We therefore retain Qwen2.5-14B as the only preregistered model-transfer evaluation. After completion of the confirmatory analyses, we additionally conducted the explicitly post-hoc Phi-4 cross-family replication described below.

Table 4: Compliance audit for candidate cross-family agents (30-question preregistered smoke evaluation; criteria fixed in advance).

Agent	**Malformed**	**Valid answer**	**Token-limit**	**Decision**
Qwen2.5-7B-Instruct	(reference)	—	—	primary agent
Mistral-7B-Instruct-v0.3	42.2%	100%	0%	excluded
Llama-3.1-8B-Instruct	5.9%	96.7%	3.3%	excluded (near boundary)
Qwen2.5-14B-Instruct	0.0%	93.3%	6.7%	**passed**

4.1 Post-hoc Cross-Family Replication with Phi-4

To test whether the observed prospective headroom is specific to the Qwen2.5 agent family, we conducted an explicitly post-hoc replication with `microsoft/phi-4` after completing the confirmatory analyses. Before examining replication outcomes, we fixed a separate post-hoc protocol-compliance screen using the same tool-use criteria. On a 50-question development smoke set, Phi-4 achieved a 92.0% valid-final rate, 0.0% malformed-search rate, 2.0% token-limit truncation rate, and zero retrieval leakage, satisfying the fixed compliance thresholds. We then evaluated Phi-4 on a representative, non-enriched HotpotQA sample of 350 questions and 860 frozen states under the same frozen-state intervention and greedy continuation protocol. The replication yields significant Oracle–CE headroom of **+0.081 F1** (95% CI [0.063, 0.102]). CE also significantly exceeds uniform random selection by **+0.048 F1** [0.029, 0.067]. Stronger relevance does not eliminate the gap: residual Oracle–BGE(q_t) headroom is **+0.071** [0.053, 0.089], and Oracle–BGE-State headroom is **+0.078** [0.060, 0.099]. Query-only BGE improves slightly over CE (+0.011 [0.002, 0.021]), whereas BGE-State does not (+0.003 [−0.010, 0.016]). Thus, stronger relevance can improve selection while substantial prospective headroom remains. Because this experiment was conducted after the confirmatory analyses, we treat it strictly as a post-hoc cross-family replication.

5 Outcome Labels and Evaluation Targets

Prospective utility and headroom. For an unseen candidate d, raw prospective interventional utility is the change in downstream answer F1 obtained by injecting d into the frozen state s_t and resuming the same policy. For a relevance selector R, headroom is

$$H_R(s_t) = Y(d_{\mathrm{PIU}}) - Y(d_R), \tag{1}$$

the difference between the best measured candidate and the selector's choice within the same intervention set $\mathcal{D}_t$. Two opportunity definitions are used and always reported separately: *Opp.-Base*, the fraction of states with at least one candidate whose raw intervention outcome strictly exceeds the base (no-intervention) continuation outcome; and *Opp.-CE*, the fraction of states with at least one candidate whose outcome strictly exceeds that of the CE-selected candidate. The substantive threshold $H \geq 0.25$ F1 distinguishes answer-level changes from marginal fluctuations.

Prospective state-identification target. The RQ3 gate predicts, before any intervention outcome is observed, the binary state-level label

$$Z(s_t) = \mathbf{1}[H_{\mathrm{CE}}(s_t) > 0]\,, \tag{2}$$

i.e., whether the current state contains any candidate that would improve on the CE selection. Gates are evaluated by AUC and average precision against this retrospectively computed label; the label itself is never available to the gate.

Post-rollout scorer labels. The exploratory continuation scorer is trained on candidate pairs whose preference labels are constructed from the raw downstream F1 outcomes of their continuations. Pairs whose outcome difference falls inside the preregistered tie zone ($\tau = 0.1$) are excluded from training; this is the same tie zone used for pairwise outcome labels. F1 is used only to construct training labels and never appears among inference-time features.

Myopic utility. For the post-hoc forced-answer analysis, myopic utility is defined analogously to PIU except that the continuation policy disables further retrieval and requires an immediate answer after candidate injection. The resulting outcome measures the candidate's immediate answer-support value under the frozen state while preventing any contribution from subsequent retrieval. This makes the utility label explicitly dependent on the continuation policy: the same candidate can carry different utility under the full policy and under the forced-answer policy.

6 Observable State, Candidate, and Continuation Features

Pre-intervention features. The interpretable feature vector $\varphi(s_t, d)$ asks what the candidate *adds* relative to current knowledge. Content tokens are lowercased alphanumeric tokens with a fixed stopword list removed; entity proxies are capitalized tokens and multi-digit numbers. With D the candidate's content-token set, $E_1, \ldots, E_k$ the per-document evidence token sets ($E = \bigcup_i E_i$), Q and Q_t the token sets of the original question and current query:

- **novelty** $= |D \setminus E|/|D|$ — fraction of candidate content unseen in evidence;
- **redundancy** $= \max_i \mathrm{Jaccard}(D, E_i)$ — maximal overlap with any single consumed document;
- **entity_gain** — new entity-proxy types per 100 tokens (scale-clipped);
- **query_align** $= |D \cap Q_t|/|Q_t|$;
- **q0_align** $= |D \cap Q|/|Q|$;
- **gap_coverage** $= |D \cap (Q \setminus E)|/|Q \setminus E|$ — coverage of question tokens not yet present in evidence;

- **doc_len** = log(1 + token count), normalized;
- **turn_idx**, **n_evidence** — search step and number of consumed documents, normalized.

Semantic variants of the alignment and novelty features (E5 embedding based) are additionally used in the gate analyses (Sec. 7). These features are used descriptively in the winner characterization and as inputs to the prospective gates; all of them are computable before any intervention outcome is observed.

Post-rollout continuation features. The exploratory continuation scorer additionally observes features of the realized continuation of each candidate: normalized CE score of the candidate; the number of additional searches in the continuation; termination indicators; trajectory length in tokens and turns; answer length and an empty-answer indicator; lexical support of the answer from the injected document; lexical support from subsequently retrieved evidence; agreement of the continuation answer with the base answer; and an indicator of whether the answer matches the modal answer among the top-3 CE candidates' continuations. These features require running the candidate's continuation and are therefore available only to the post-rollout scorer, never to the pre-intervention gates.

7 Additional Diagnostic Results

7.1 Where Utility Winners Appear in the Relevance Ranking

Table 5 characterizes, among CE-relative opportunity states, the CE-rank position of utility-maximizing candidates. Winners concentrate near, but not at, the top of the relevance ranking.

Tie handling for winner-rank characterization. When multiple candidates attain the same maximal downstream outcome within a state, we fractionally distribute that state's winner mass across all tied maximizers, analogously to the candidate-family attribution analysis. Top-k shares are therefore computed from tie-weighted winner mass, and median ranks from the corresponding weighted empirical distribution.

Table 5: Tie-aware CE-rank position of utility-maximizing candidates among CE-relative opportunity states. When multiple candidates attain the same maximal downstream outcome, each receives equal fractional winner mass within the state. Top-2/Top-3 report tie-weighted winner mass within CE rank 2/3; median ranks are computed from the corresponding weighted empirical distribution, and the final column reports winner mass at CE rank ≥ 5.

Setting	Opp. states	Median CE rank	Top-2	Top-3	Rank 5+
HotpotQA dev	81	4	17.3%	38.3%	34
HotpotQA lockbox	129	3	33.3%	50.4%	42
2Wiki	164	4	23.8%	47.0%	58
Qwen2.5-14B	140	4	30.7%	45.7%	54
MuSiQue	187	4	17.1%	35.3%	84

Outcome-permutation characterization. To describe which observable properties distinguish winning candidates, we compare each feature's observed winner statistic against a within-state outcome-permutation null. This analysis is exploratory and descriptive; the reported directions are

associations under the tested null, not causal effects. Table 6 summarizes how many of the five settings fall outside the permutation null for each feature.

Table 6: Within-state outcome-permutation characterization of winning candidates (exploratory). "Direction" describes the observed tendency in the settings outside the null.

Feature	**Settings outside null**	**Direction**
Document length	5/5	higher
Current-query alignment	4/5	higher
Unresolved-term coverage	4/5	higher
q_0 alignment	3/5	higher
Redundancy	3/5	lower
Novelty	2/5	mixed / limited
Entity gain	0/5	no consistent signal

7.2 Detailed Prospective State-Identification Results

The RQ3 gate predicts $Z(s_t) = \mathbf{1}[H_{\mathrm{CE}}(s_t) > 0]$ from pre-intervention information only: CE margin and CE score statistics, candidate-aggregate features, and probe signals. Table 7 reports discrimination of the full shallow gate; across post-hoc gate families, the best per-setting AUC ranges from 0.567 to 0.637, and CE margin or CE confidence alone performs approximately at chance.

Table 7: Prospective state-identification AUC of the full shallow gate against $Z(s_t)$, by evaluation setting. Across post-hoc gate families, the best per-setting AUC ranges from 0.567 to 0.637; for Qwen2.5-14B the best family is the candidate-aggregate gate (0.568) rather than the full gate.

Setting	**Full shallow gate AUC**
HotpotQA dev	0.599
HotpotQA lockbox	0.637
2Wiki	0.527
Qwen2.5-14B	0.560
MuSiQue	0.505

Table 8: Gate-family comparison on the HotpotQA lockbox (prevalence of $Z(s_t) = 1$: 0.188). The full shallow gate attains an average-precision lift of 1.49 relative to prevalence.

Gate family	**Lockbox AUC**
CE margin	0.479
CE score statistics	0.494
Candidate aggregates (φ)	0.547
MiniLM (fine-tuned)	0.500
DeBERTa (fine-tuned)	0.525
Full shallow gate	0.637

7.3 Budgeted Lookahead and Simple Aggregation

Table 9 reports the fraction of oracle headroom recovered when the intervention budget is limited to the top-k CE-ranked candidates.

Table 9: Fraction of Oracle–CE headroom recovered under a top-k CE-ranked intervention budget.

Setting	Oracle@2 recovery	Oracle@3 recovery
HotpotQA dev	30.9%	64.0%
HotpotQA lockbox	46.4%	65.4%
2Wiki	37.3%	60.8%
Qwen2.5-14B	46.3%	70.6%
MuSiQue	32.2%	54.2%

Recovery continues to rise with budget: 68.2–80.0% of oracle headroom at $k = 4$ and 85.8–95.2% at $k = 6$ across the five settings. Simple aggregation over the rolled-out candidates does not convert this budgeted signal into a selector: majority voting over top-k continuation answers degrades performance relative to CE selection in every setting (negative recovery, -5.9% to -58.0% at $k \in \{4, 6\}$), and early-termination heuristics are inconsistent across settings. One reason is answer modality: the useful oracle candidate's continuation answer matches the modal answer among top-3 continuations in only 22.5–28.7% of cases, whereas the CE candidate's answer is modal in 34.1–47.9%; useful answers are disproportionately non-modal.

7.4 Exploratory Post-Rollout Continuation Scorer

An unconstrained selector was first evaluated and did not improve over CE on HotpotQA development (development gain: -0.022 F1). We therefore treat this variant as unsuccessful rather than selecting it for transfer evaluation. The reported variant is CE-sticky: it overrides the CE choice only when the scorer margin exceeds a threshold selected on HotpotQA development and then frozen for all non-development evaluations. This is a post-rollout reranker requiring up to k continuation rollouts per state; it is not a pre-intervention selector, and its results are explicitly exploratory.

Table 10: CE-sticky post-rollout continuation scorer (exploratory). Overrides counts states where the scorer replaces the CE choice.

Setting	Overrides	Gain vs. CE	95% CI	Oracle@3 recovered
HotpotQA dev	12/579	+0.0011	[−0.003, 0.005]	2.1%
HotpotQA lockbox	13/686	+0.0035	[0.0007, 0.0071]	4.6%
2Wiki	39/819	+0.0075	[0.0017, 0.0143]	8.7%
Qwen2.5-14B	49/838	+0.0096	[0.0025, 0.0172]	14.0%
MuSiQue	58/959	+0.0004	[−0.004, 0.005]	0.8%

The continuation scorer is fit only on the designated training split; the CE-sticky margin is selected on HotpotQA development and then frozen for all non-development evaluations.

7.5 Trajectory Decomposition

The main paper reports the lockbox decomposition in detail together with the query-change stratum for the remaining settings; the full per-setting table (strata: no search / all / any search / both search / query changed, with state counts, headroom confidence intervals, and Opp.-CE) is provided with the frozen analysis reports in the repository. On the lockbox, the query-changed stratum ($n = 175$) shows headroom of $+0.311$ $[0.244, 0.380]$ with Opp.-CE of 44.6%. The decomposition is computed over the post-intervention behavior of the two compared arms (d_{PIU} and d_{CE}); its

denominator therefore differs from the all-arm average of 0.62 additional search steps reported in Sec. 1. Post-intervention strata are descriptive and should not be interpreted causally.

7.6 Myopic Utility versus Full Prospective Utility

The trajectory decomposition above is descriptive because its strata are defined after intervention. We therefore conduct a complementary forced-answer analysis on the HotpotQA lockbox. For each frozen state and candidate, the full-PIU arm injects the candidate and resumes the normal greedy continuation policy, whereas the myopic arm injects the same candidate but disables further retrieval and requires an immediate answer. All other state, candidate, and evaluation conditions are held fixed. Retrieval-free myopic utility is moderately correlated with full PIU (candidate-level Spearman $\rho = 0.652$), but its within-state ordering is much coarser: mean within-state Spearman correlation is 0.392 and the myopic maximum is tied in 84.1% of states. Among CE-relative opportunity states, strict winner agreement with full PIU is only 27.1%. The myopic oracle still exhibits substantial headroom (+0.083 F1, 95% CI [0.063, 0.104]), showing that immediate answer effects are real. However, when ties are resolved in favor of CE, myopic selection gains only +0.015 F1 [0.003, 0.027] on full-continuation outcomes, recovering **12.7%** of the full Oracle–CE headroom. Thus, immediate answer improvement alone is insufficient to identify which evidence is most valuable under continued search.

Table 11: Post-hoc forced-answer comparison of myopic utility and full PIU on the HotpotQA lockbox.

Quantity	**Result**
Candidate-level Spearman ρ	.652
Mean within-state Spearman	.392
States with tied myopic maximum	84.1%
Strict winner agreement (opportunity states)	27.1%
Full Oracle–CE	+.117 [.092, .142]
Myopic Oracle–CE	+.083 [.063, .104]
Myopic selection on full outcomes	+.015 [.003, .027]
Full headroom recovered	12.7%

8 Robustness and Sensitivity Results

Exact random selection and representative strata. The random baseline is computed in closed form as the exact expectation of a uniform draw over $\mathcal{D}_t$ (the uniform average of the candidate outcomes), and compared with CE selection using paired question-level confidence intervals. CE selection is significantly better than this exact random baseline in all five settings (CE − Random from +0.0228 to +0.0720 F1), so the reported headroom is measured beyond a genuinely informative baseline. On the representative non-enriched strata, the Oracle–CE effect persists: question-level estimates range from +0.062 to +0.159 F1 across the five settings, with every confidence interval excluding zero. Failure enrichment therefore does not create the Oracle–CE effect; it only concentrates evaluation on states where selection matters.

Opportunity magnitude. Table 12 decomposes both opportunity definitions by effect threshold. By definition, Opp.-Base $\geq$ Opp.-CE within every row, and > 0 rates bound the $\geq .25$ rates within each definition.

Table 12: Opportunity rates under both definitions and effect thresholds.

Setting	Opp.-Base > 0	Opp.-Base $\geq .25$	Opp.-CE > 0	Opp.-CE $\geq .25$
HotpotQA dev	30.1%	26.4%	14.0%	12.4%
HotpotQA lockbox	29.3%	27.8%	18.8%	16.9%
2Wiki	26.4%	24.8%	20.0%	17.6%
Qwen2.5-14B	32.1%	29.1%	16.7%	14.3%
MuSiQue	21.8%	18.8%	19.5%	16.1%

Noise, replay, and cross-fitted robustness. Label reliability on the replay-audited subset is 97.8–99.0% (three greedy continuations per intervention on 20% of states; 115/579 primary states and 137/686 lockbox states). A fresh 150-question cross-fitted experiment separates candidate selection from outcome evaluation using two independent rollout sets, yielding an Oracle–CE estimate of +0.111 F1 (95% CI $[0.080, 0.144]$), nearly identical to the naive +0.114 estimate on the same sample (A→B: +0.117; B→A: +0.104).

Post-hoc sampled-decoding policy sensitivity. To avoid selecting and evaluating the oracle candidate on the same stochastic realization, we use held-out replicate evaluation. For each fold, candidate selection is performed using the remaining sampled replicates and the selected candidate is evaluated on the held-out replicate. Across the full 686-state lockbox, the three held-out Oracle–CE estimates are +0.0298, +0.0281, and +0.0407 F1. Thus, realized PIU remains positive under sampled continuation but is substantially smaller than under the primary greedy policy, demonstrating policy dependence. The sampled-decoding runs use $T = 0.7$, top-$p = 1.0$, with fixed seeds 101, 102, and 103; prompts, stop sequences, parsing, retrieval, and search-budget logic are otherwise unchanged.

Matched intervention controls. The selection-specific controlled contrast is the primary control estimand: for the candidates actually selected by CE and by the PIU oracle, Table 13 reports each selection's outcome relative to its matched control and the resulting controlled Oracle–CE contrast.

Table 13: Selection-specific matched-control contrasts (F1).

Setting	CE − control	Oracle − control	Controlled Oracle−CE
HotpotQA dev	+0.085	+0.160	+0.075
HotpotQA lockbox	+0.044	+0.152	+0.109
2Wiki	+0.025	+0.164	+0.138
Qwen2.5-14B	+0.066	+0.162	+0.097
MuSiQue	+0.030	+0.113	+0.083

The paired controlled Oracle−CE contrast excludes zero in all five settings (e.g., HotpotQA dev +0.0748 $[0.0516, 0.0992]$; 2Wiki +0.1383 $[0.1103, 0.1678]$). The near-zero all-arm average is not the primary selection-specific estimand: many intervention candidates are never selected by either CE or the oracle. For the candidates actually selected by CE and by the PIU oracle, the controlled contrast remains positive in all five settings. On the lockbox, the all-arm controlled average is +0.0060 F1 with a 95% CI of $[-0.0017, 0.0137]$; excluding controls already present in the agent context changes it only slightly (+0.0060 to +0.0051; exclusion CI $[-0.0027, 0.0127]$), and

for interventions assigned a second independently sampled control, effect signs agree in 87–91% of cases.

Additional matching-specification sensitivity. As a post-hoc exploratory robustness analysis, we reran all 5,487 lockbox interventions using three relaxed control-matching specifications: removing the BM25 rank-bucket constraint, removing the document-length constraint, or removing both. The resulting all-arm controlled estimates are +0.0099, +0.0089, and +0.0086 F1, respectively, compared with +0.0060 under the full confirmatory specification (Table 14); relaxing the matching constraints therefore does not attenuate the diagnostic. Table 15 verifies that the relaxed specifications actually change the sampled controls.

Table 14: Post-hoc sensitivity to matched-control specifications on the HotpotQA lockbox. Estimates are question-level means with question-bootstrap 95% confidence intervals.

Matching specification	**Mean ΔF1**	**95% CI**
Full confirmatory	+0.0060	[−0.0017, 0.0137]
No rank bucket	+0.0099	[0.0024, 0.0177]
No length	+0.0089	[0.0009, 0.0171]
Neither	+0.0086	[0.0006, 0.0167]

Table 15: Manipulation check for relaxed control-matching specifications.

Specification	**Rank-bucket match**	**Length within ±25%**	**Fallback**
No rank bucket	11.0%	97.0%	0.3%
No length	59.4%	22.3%	1.8%
Neither	9.8%	20.6%	0.1%

Oracle tie handling. The intervention pipeline resolves equal maxima by the fixed candidate enumeration order `bm25_top` → `dense_top` → `ce_top` → `mid_rel` → `novelty` → `redundancy` → `entity_bridge` → `random`. On the HotpotQA lockbox, 580 of 686 states have multiple candidates tied at the maximum raw utility; most of these ties occur because the maximum utility is zero. Among states with a positive maximum, 100 states (14.6% of all lockbox states) contain a tied maximizer. This tie rule cannot change Oracle–CE headroom because tied candidates have identical downstream F1 within the state, but it can affect descriptive candidate attribution and winner-rank statistics. We therefore use fractional tie weighting for both candidate-family attribution and the winner-rank characterization (Sec. 7.1). For this reason, family shares below divide oracle-winning mass fractionally across tied maximizers.

Candidate-family attribution. Because multiple candidates can attain the same maximal downstream outcome, assigning each state to the first enumerated maximizer biases family counts. We therefore fractionally divide each state's oracle-winning mass across all tied maximizers. Under this tie-split attribution, the three standard-retrieval candidate families (`bm25_top`, `dense_top`, and `ce_top`) account for 56.8% of oracle-winning mass, while the five auxiliary families account for 43.2%. Table 16 reports the per-family breakdown, and Table 17 extends the pool-sensitivity analysis to all five evaluation settings.

Table 16: Candidate-family attribution on the HotpotQA lockbox (686 states; 201 states with a positive opportunity). Oracle-winning mass is fractionally divided across candidates tied at the maximum downstream outcome. The final column applies the pipeline's fixed first-maximum tie rule and is shown only to illustrate enumeration bias.

Family	**Mean U**	**Tie-split share**	**First-max share**
`bm25_top`	+0.0180	8.9%	25.4%
`dense_top`	+0.0638	25.7%	33.3%
`ce_top`	+0.0652	22.2%	21.9%
`mid_rel`	+0.0182	7.1%	3.0%
`novelty`	+0.0239	8.7%	6.0%
`redundancy`	+0.0172	9.0%	4.0%
`entity_bridge`	+0.0210	9.4%	4.5%
`random`	+0.0255	8.9%	2.0%
Standard retrieval	—	56.8%	—
Auxiliary families	—	43.2%	—

Table 17: Candidate-pool sensitivity across all five evaluation settings. Standard and auxiliary columns show tie-split oracle-winning mass. Standard-only headroom restricts the oracle to BM25, dense-retrieval, and CE candidates. Bracketed values are question-level 95% confidence intervals.

Setting	**Standard**	**Aux.**	**Full headroom**	**Standard-only**
HotpotQA dev	60.0%	40.0%	+0.082	+0.042 [0.029, 0.058]
HotpotQA lockbox	56.8%	43.2%	+0.117	+0.078 [0.059, 0.100]
2Wiki	50.4%	49.6%	+0.142	+0.083 [0.064, 0.104]
Qwen2.5-14B	61.8%	38.2%	+0.097	+0.071 [0.053, 0.091]
MuSiQue	42.0%	58.0%	+0.097	+0.047 [0.035, 0.060]

Restricting the intervention pool to standard retrieval candidates leaves positive Oracle–CE headroom in all five settings, with every question-level 95% confidence interval excluding zero. Auxiliary candidates are not inert: they account for 38–58% of tie-split oracle-winning mass across settings, with the largest share on MuSiQue.

Cost accounting. 96.2% of rollout token cost is prompt tokens (context replay), implying that amortized approximations of prospective intervention should target context reuse rather than generation length.

9 Reproducibility

Software and hardware. vLLM 0.10.2, PyTorch 2.8.0 (CUDA 12.8), Transformers 4.55.2, `bm25s` 0.3.10, scikit-learn 1.9.0, Python 3.11; NVIDIA H100 80GB. Confirmatory evaluations use greedy decoding; fixed seeds are used for sampling, stratification, and control matching. The post-hoc sampled-decoding sensitivity analysis uses $T = 0.7$, top-$p = 1.0$, with fixed seeds 101, 102, and 103.

Scale of the study. Per-setting confirmatory sizes (questions / frozen states): HotpotQA dev 297 / 579; lockbox 350 / 686; 2Wiki 350 / 819; Qwen2.5-14B 350 / 838; MuSiQue 349 / 959. The cross-fitted oracle uses 150 fresh questions with two independent rollout sets. The post-hoc choice-space expansion adds 3,861 full-pool BGE/BGE-State intervention arms. The sampled-decoding

sensitivity evaluates 6,173 lockbox base/intervention arms under three independent stochastic replicates. The post-hoc Phi-4 cross-family replication uses a representative, non-enriched 350-question HotpotQA sample containing 860 frozen states and 6,870 intervention arms. The forced-answer myopic analysis evaluates 6,173 lockbox arms (686 base continuations and 5,487 candidate interventions), with retrieval disabled after candidate injection.

Analysis provenance. The continuation scorer is fit only on the designated training split; the CE-sticky margin is selected on HotpotQA development and then frozen for all non-development evaluations. All post-hoc analyses (choice-space expansion, winner characterization, lookahead, scorer, sampled decoding, Phi-4 cross-family replication, myopic-utility analysis, and matching sensitivity) are labeled as such in the frozen reports.

Code and records. Anonymized code, the preregistration document, and frozen analysis reports are available at the repository linked in the abstract. All confirmatory numbers in the main paper trace to a preregistered criterion and a frozen report file in that repository.